GNSS Signal Engineering

Physical Layer Dynamics, Signal Structure, and Stochastic Estimation in Global Positioning Systems.

The GNSS Landscape

GPS (United States)

First Launch: 1978. Fully Operational: 1995. The baseline constellation for global L-band positioning.

GLONASS (Russia)

First Launch: 1982. Fully Operational: 1995. Unique FDMA architecture across multiple frequencies.

Galileo (European Union)

First Launch: 2011. Operational: 2016. High-integrity civilian signal utilizing BOC modulation.

BeiDou (China)

First Launch: 2000 (Regional). Global (BDS-3): 2020. Features high-inclination geostationary components.

The Space Segment: MEO Dynamics

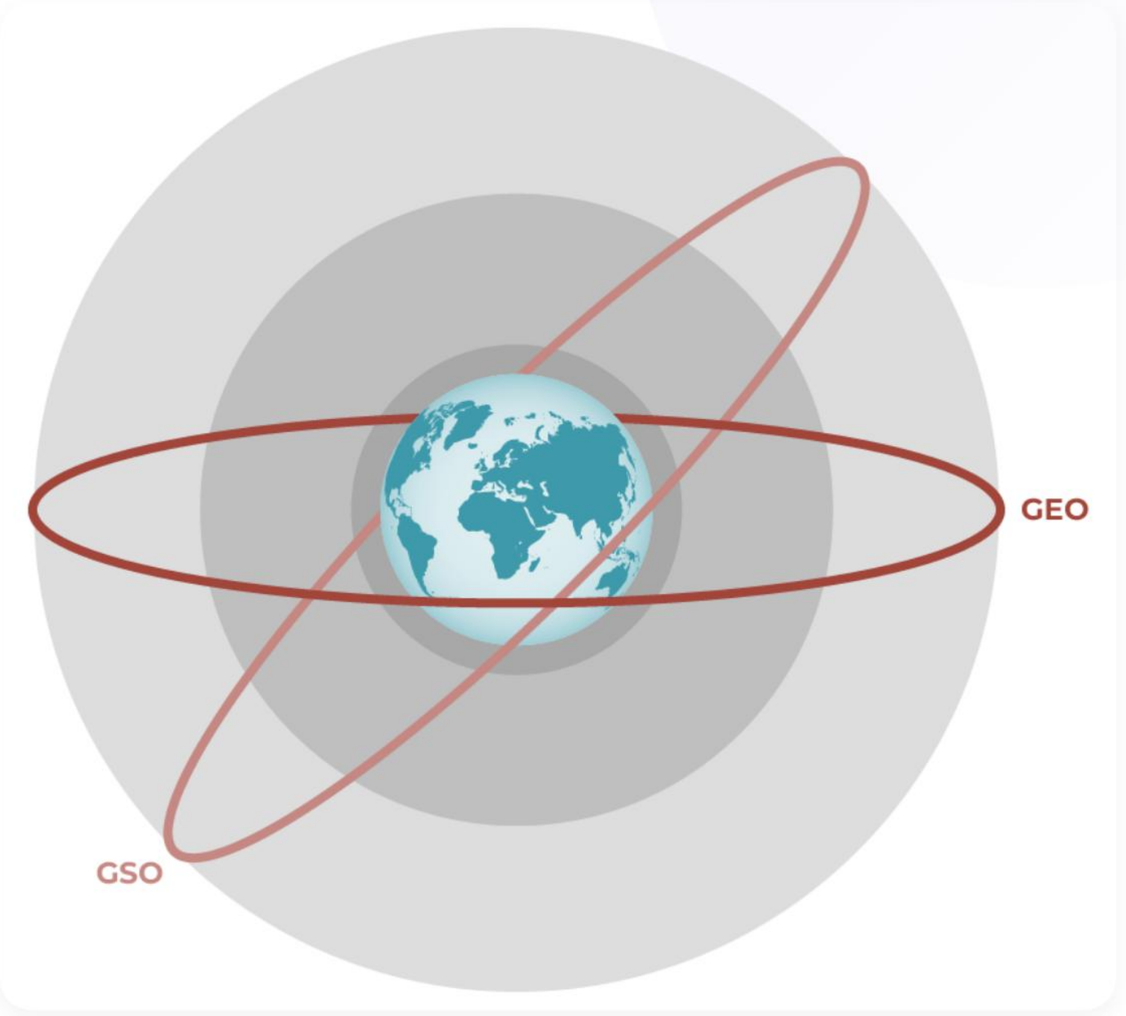
Orbital Mechanics

- **Altitude:** ~20,200 km (MEO - Medium Earth Orbit).
- **Orbital Period:** ~12 Sidereal Hours.
- **Speed:** ~3.9 km/s.

Signal Propagation

Speed of light (c): 299,792.458 km/s.

$$T_{transit} \approx 67ms \text{ to} 85ms$$



Signal Modulation Techniques

CDMA

Code Division Multiple Access allows multiple satellites to share the same frequency band via orthogonal codes.

BPSK

Binary Phase Shift Keying. Legacy modulation for GPS L1 C/A (180° phase reversals).

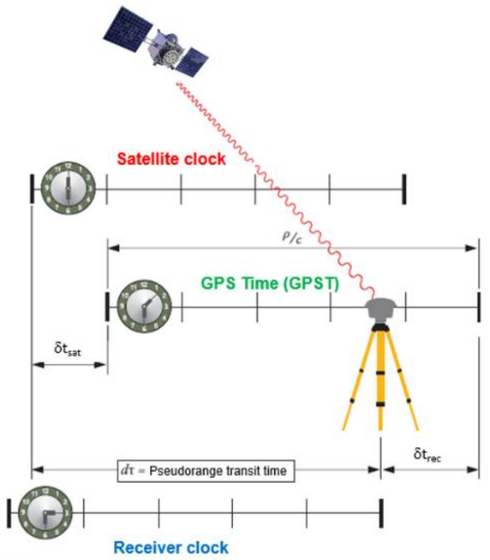
BOC

Binary Offset Carrier. Splitting the signal energy to sharpen the correlation peak and mitigate multipath.

PRN Code Architecture

Signal Type	Code Length (Chips)	Chipping Rate	Repetition Period
GPS L1 C/A (Civil)	1,023	1.023 MHz	1 Millisecond
GPS P(Y) (Military)	~6.18 Trillion	10.23 MHz	1 Week
Galileo E1 (Modern)	4,092	1.023 MHz	4 Milliseconds

Pseudorange Determination



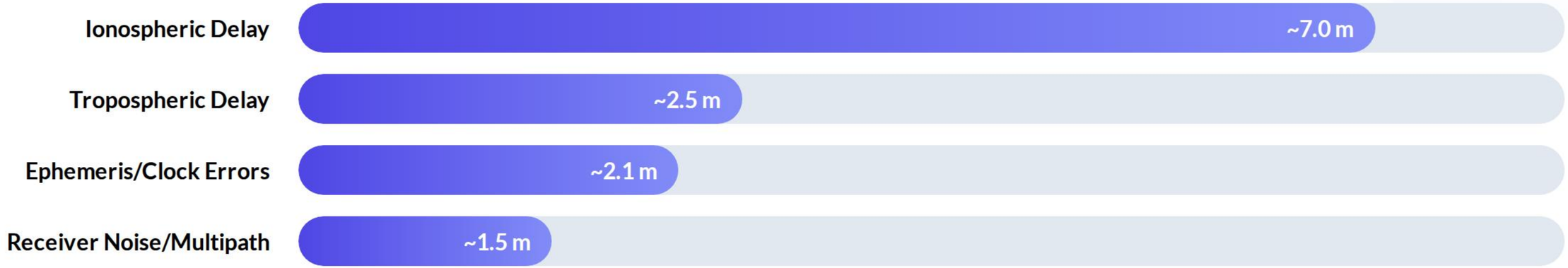
Calculating ToA

The receiver correlates the incoming signal with a local replica. The offset required to align the PRN codes determines the flight time.

$$\rho = c \cdot t_{rx} - t_{tx}$$

The measurement is "Pseudo" because of un-synchronized receiver and satellite clocks.

The Stochastic Error Budget



Relativistic Bias: Clocks gain ~38 microseconds per day due to gravitational potential and velocity differences.

The Navigation State Problem

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Unknowns

Solving for (x, y, z, dt)

We solve the system of non-linear equations for the receiver's 3D coordinates and its internal clock bias.

$$\rho_i = \sqrt{x_i - x^2 + y_i - y^2 + z_i - z^2} + c \cdot dt$$

Estimation: MLE vs. NLS

MLE Foundation

Maximum Likelihood Estimation maximizes the probability of observing measurements under Gaussian noise assumptions. In GNSS, MLE and LS are equivalent for white noise.

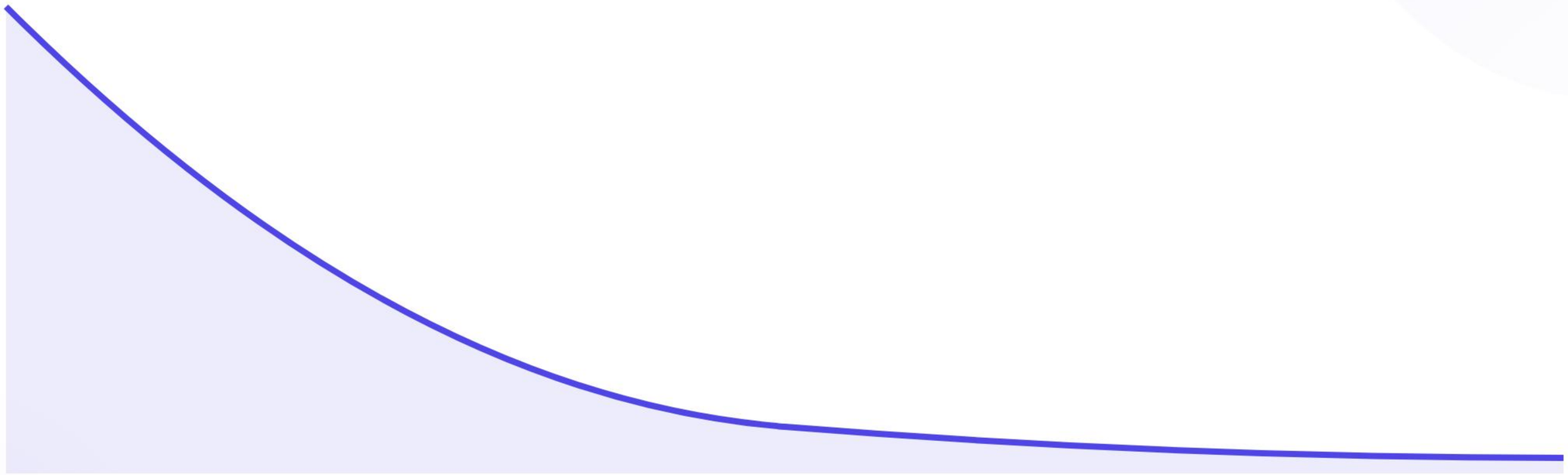
NLS Linearization

Non-Linear Least Squares uses Taylor expansion to linearize the range equations around an initial guess, iterating until convergence.

$$G\Delta x = \Delta\rho$$

Iterative WNLS Refinement

Residual Convergence (RMSE)



Weighted Non-Linear Least Squares: Accounts for signal quality (SNR) and satellite elevation by applying a weight matrix W.

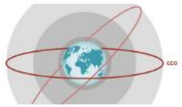
$$\Delta x = G^T W G^{-1} G^T W \Delta \rho$$

Technical Summary

A successful GNSS position solution depends on the precision of the local PRN correlation, the accuracy of the atmospheric delay models, and the robustness of the iterative estimator.

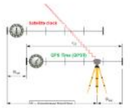
Questions?

Image Sources



https://cdn.sparkfun.com/assets/home_page_posts/4/2/7/8/gso-vs-geo.png

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